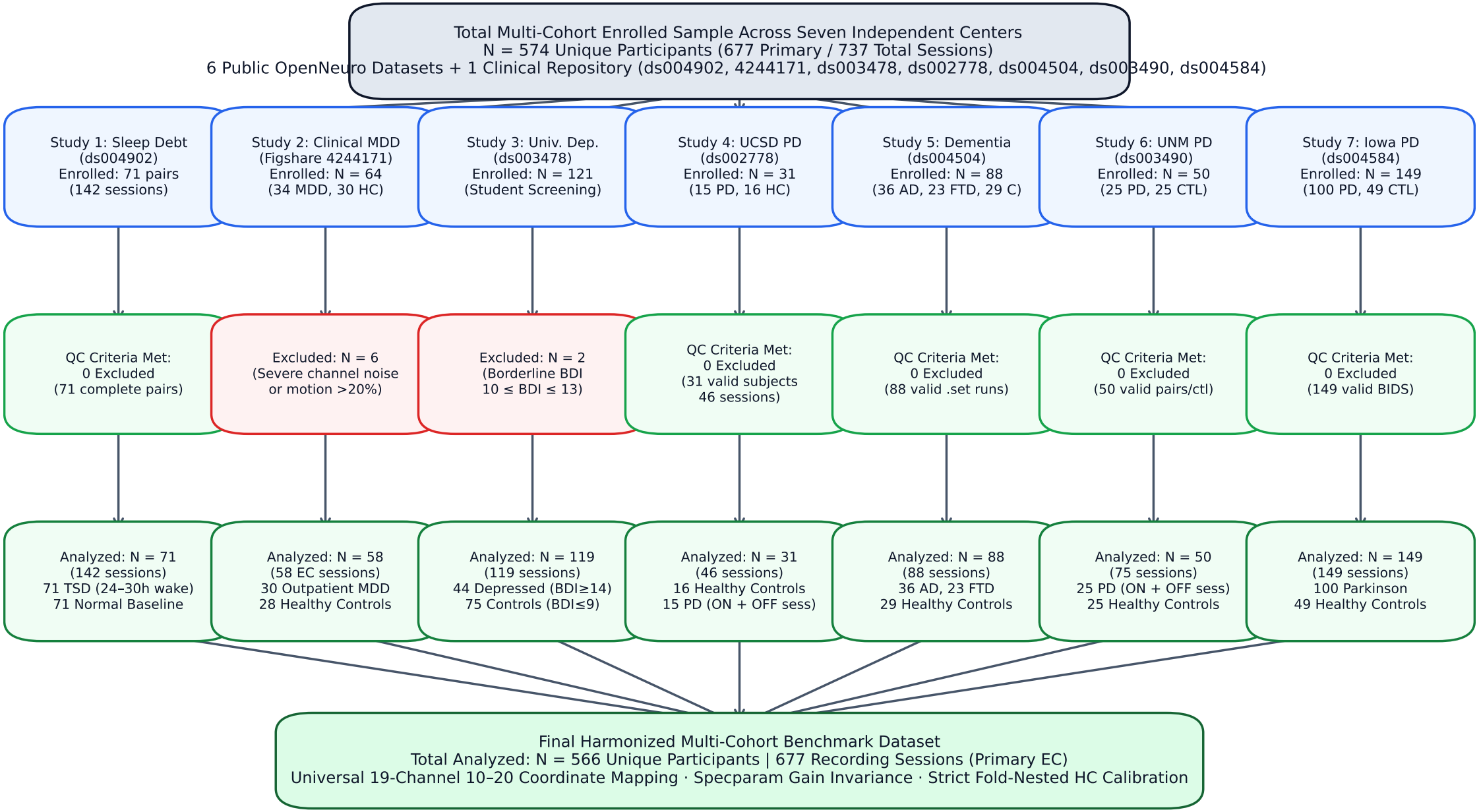
Total Multi-Cohort Enrolled Sample Across Seven Independent Centers
N = 574 Unique Participants (677 Primary / 737 Total Sessions)
6 Public OpenNeuro Datasets + 1 Clinical Repository (ds004902, 4244171, ds003478, ds002778, ds004504, ds003490, ds004584)
Study 1: Sleep Debt (ds004902) Enrolled: 71 pairs (142 sessions)
Study 2: Clinical MDD (Figshare 4244171) Enrolled: N = 64 (34 MDD, 30 HC)
Study 3: Univ. Dep. (ds003478) Enrolled: N = 121 (Student Screening)
Study 4: UCSD PD (ds002778) Enrolled: N = 31 (15 PD, 16 HC)
Study 5: Dementia (ds004504) Enrolled: N = 88 (36 AD, 23 FTD, 29 C)
Study 6: UNM PD (ds003490) Enrolled: N = 50 (25 PD, 25 CTL)
Study 7: Iowa PD (ds004584) Enrolled: N = 149 (100 PD, 49 CTL)
QC Criteria Met: 0 Excluded (71 complete pairs)
Excluded: N = 6 (Severe channel noise or motion >20%)
Excluded: N = 2 (Borderline BDI 10 ≤ BDI ≤ 13)
QC Criteria Met: 0 Excluded (31 valid subjects 46 sessions)
QC Criteria Met: 0 Excluded (88 valid .set runs)
QC Criteria Met: 0 Excluded (50 valid pairs/ctl)
QC Criteria Met: 0 Excluded (149 valid BIDS)
Analyzed: N = 71 (142 sessions) 71 TSD (24–30h wake) 71 Normal Baseline
Analyzed: N = 58 (58 EC sessions) 30 Outpatient MDD 28 Healthy Controls
Analyzed: N = 119 (119 sessions) 44 Depressed (BDI≥14) 75 Controls (BDI≤9)
Analyzed: N = 31 (46 sessions) 16 Healthy Controls 15 PD (ON + OFF sess)
Analyzed: N = 88 (88 sessions) 36 AD, 23 FTD 29 Healthy Controls
Analyzed: N = 50 (75 sessions) 25 PD (ON + OFF sess) 25 Healthy Controls
Analyzed: N = 149 (149 sessions) 100 Parkinson 49 Healthy Controls
Final Harmonized Multi-Cohort Benchmark Dataset
Total Analyzed: N = 566 Unique Participants | 677 Recording Sessions (Primary EC)
Universal 19-Channel 10–20 Coordinate Mapping · Specparam Gain Invariance · Strict Fold-Nested HC Calibration